

Analysis of Scientific Data



Lecture 6

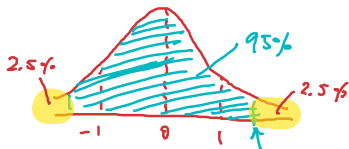
Making Inferences

Poll Question

Suppose a random variable has a Normal distribution.

How many standard deviations from the mean cover 95% of possible outcomes?

1. 1.282
2. 1.645
3. 1.960
4. 2.054



$$qnorm(q) \quad P(Z < \text{yellow circle}) = q$$
$$qnorm(0.975)$$

Confidence Interval

Recall that for large n ,

CLT. $\bar{X} \underset{\sim}{\text{approx}} \text{Normal}\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \underset{\sim}{\text{approx}} \text{Normal}(0, 1)$$

$$0.95 = \mathcal{P}(-1.96 < Z < 1.96)$$

$$0.95 = \mathcal{P}\left(-1.96 < \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right)$$

$$= \mathcal{P}\left(-1.96 \frac{\sigma}{\sqrt{n}} < \bar{X} - \mu < 1.96 \frac{\sigma}{\sqrt{n}}\right)$$

$$= \mathcal{P}\left(-1.96 \frac{\sigma}{\sqrt{n}} < \mu - \bar{X} < 1.96 \frac{\sigma}{\sqrt{n}}\right)$$

$$0.95 = \mathcal{P}\left(\bar{X} - 1.96 \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96 \frac{\sigma}{\sqrt{n}}\right)$$

Confidence Interval

We say we are 95% *confident* that the population mean is

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}},$$

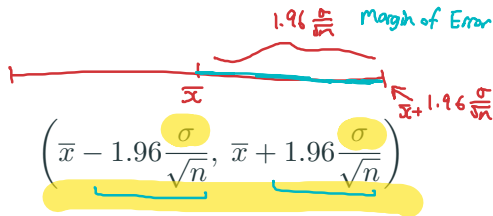
or that it is in the interval

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right).$$

We call this range a 95% *confidence interval* for the population mean. μ

However we cannot actually calculate this interval... ☹

Poll Question



If we sample from a population which is more variable (σ is larger) then a confidence interval for the population mean would be

1. narrower
2. wider
3. not sure

Poll Question

$$\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$$

↖

Increasing the sample size would give a confidence interval for the population mean that is

1. narrower
2. wider
3. not sure

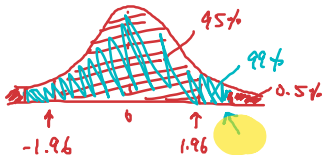
Poll Question

$$\left(\bar{x} - \underline{1.96} \frac{\sigma}{\sqrt{n}}, \bar{x} + \underline{1.96} \frac{\sigma}{\sqrt{n}} \right)$$

default 95%

Compared to a 95% confidence interval, a 99% confidence interval for the population mean would be

1. narrower
2. wider
3. not sure



$$qnorm(0.995) = 2.58$$

Poll Question

$$\frac{2.58}{1.96} \approx 1.31$$

$\Rightarrow$ about 31%
wider

How much wider is a 99% confidence interval than a 95% confidence interval?

1. About 4% wider
2. About 33% wider
3. About twice as wide
4. Infinitely wider

Poll Question

How much wider is a 100% confidence interval than a 95% confidence interval?

1. About 5% wider
2. About 33% wider
3. About twice as wide
4. Infinitely wider



Standard Error

We have seen that

$$\text{sd}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

real sd

In practice, we usually do not know the value of σ so we have to estimate it using the sample standard deviation s .

This gives the estimated standard deviation of the sample mean, known as the standard error,

$$\text{se}(\bar{x}) = \frac{s}{\sqrt{n}}$$

estimated sd

This comes from a particular sample, indicated by the lowercase $\bar{x}$.

The standardised random variable

Z-statistic

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \approx \text{Normal}(0, 1).$$

If we don't know σ and have to use the sample standard deviation S instead, this becomes the T statistic

$$T_{n-1} = \frac{\bar{X} - \mu}{S / \sqrt{n}}.$$

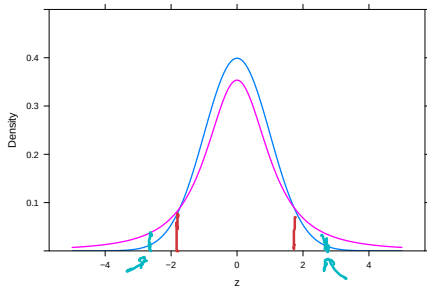
This random variable has **Student's t distribution** with $n - 1$ **degrees of freedom**, T_{n-1} .

Poll Question

The density curve for the T_2 distribution is

1. Blue

2. Pink



Confidence Intervals

Putting this in practice, a 95% confidence interval for a population mean μ is

$$\bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}},$$

where t_{n-1}^* is the critical value such that

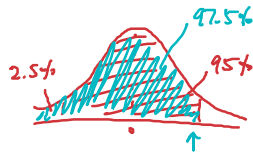
$$P(T_{n-1} \leq t_{n-1}^*) = 0.975.$$

E.g. $qt(0.975, df=2) = 4.3$

The quantity $t_{n-1}^* \frac{s}{\sqrt{n}}$ is called the (95%) margin of error.

If $n=25$, $t_{24}^* = 2.06$ 1.96

As $n \rightarrow \infty$, T_{n-1} approach the normal distribution. limiting distribution



Example: Mean Travel Time

Estimate mean travel time based on a sample of 25 students.

$$\bar{x} = 40.48, \quad s = 33.84, \quad t_{24}^* = 2.06$$

$$\begin{aligned} & \bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}} \\ &= 40.48 \pm 2.06 \times \frac{33.84}{\sqrt{25}} \\ &= \underline{40.48 \pm 13.94} \end{aligned}$$

$$\text{or } = (26.54, 54.42)$$

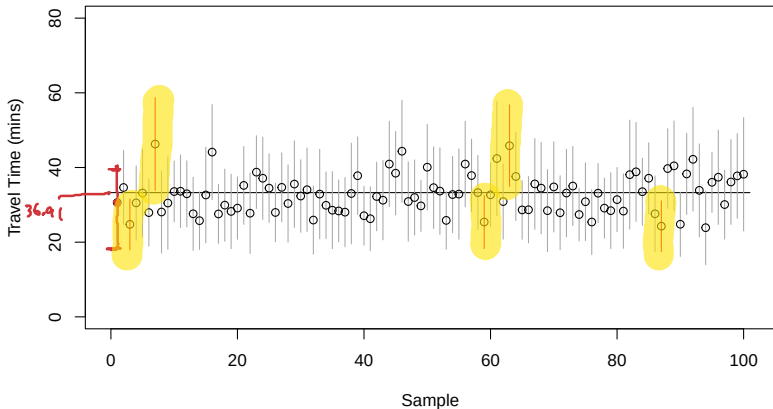
We're 95% confident that the mean travel time is between 26.54 and 54.42 minutes.

Recall that the true mean travel time is $\mu = 36.91$ mins.

Was the confidence interval we just calculated "successful"? **yes**

Confidence Intervals ($n = 25$)

Suppose we repeated this process 100 times. How many confidence intervals would contain the mean?



Poll Question

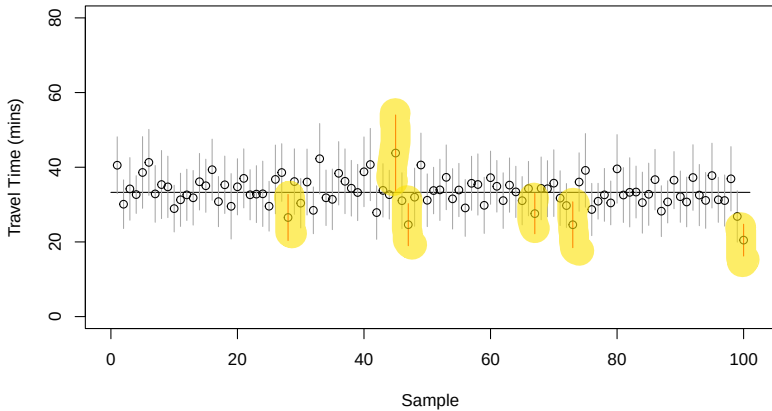
If we increase the sample size from $n = 25$ to $n = 50$, the number of 95% confidence intervals that *do not* contain the mean will be

1. higher
2. lower
3. about the same

Confidence Intervals ($n = 50$)

$$\bar{x} \pm \underbrace{1.96 \frac{\sigma}{\sqrt{n}}}_{\text{h.e.}}$$

$$\bar{x} \pm t_{n-1}^* \frac{s}{\sqrt{n}}$$



Poll Question

$$SE(\bar{x}) = \frac{s}{\sqrt{n}} \quad , \quad SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

What is the difference between a standard deviation (SD) and a standard error (SE)?

1. SE applies to a sample statistic, SD to a random variable
2. SD applies to a sample statistic, SE to a random variable
3. SE applies to a population, SD to an individual
4. They are synonymous

Week 4 Dice Rolls

X = number of attempts to pass the driving test

$$p = \frac{1}{3}$$

$$E(X) = 3$$

We can also use the t distribution to calculate the p -value for a hypothesis test.

For example, in the Week 4 applied class everyone rolled a die until they obtained a 5 or 6, recording the number of attempts.

Did the observations from this process match the theory?

Poll Question

Let μ be the mean number of attempts to obtain a 5 or 6 in the applied class setting.

If we wanted to detect any bias in this process we should test

- ~~1. $H_0 : \mu = 0$ versus $H_1 : \mu > 0$~~
2. $H_0 : \mu = 3$ versus $H_1 : \mu > 3$
3. $H_0 : \mu = 3$ versus $H_1 : \mu < 3$
4. $H_0 : \mu = 3$ versus $H_1 : \mu \neq 3$

Week 4 Dice Rolls

$$n = 703$$

The mean of the 703 attempts in Week 4 was $\bar{x} = 3.24$ with standard deviation $s = \underline{2.45}$.

Is there any evidence that the applied class process was biased?

$$H_0: \mu = 3, \quad H_1: \mu \neq 3$$

one-sided

$$p\text{-value} = P(\bar{X} \geq 3.24 \mid H_0 \text{ true})$$

$$= P\left(\frac{\bar{X} - \mu}{s/\sqrt{n}} \geq \frac{3.24 - \mu}{s/\sqrt{n}} \mid \mu = 3\right)$$

$$= P\left(T_{n-1} \geq \frac{3.24 - 3}{2.45/\sqrt{703}}\right)$$

$$= P(T_{702} \geq 2.5973)$$

$$= 0.0048$$



$$\text{Two-sided } p\text{-value} = 2 P(\bar{X} \geq 3.24 \mid H_0 \text{ true}) = 2 \times 0.0048 = 0.0096$$

$\therefore$ Reject H_0 .

One-sample t -test

More details on UQ Extend. We only summarise the testing procedure below.

- Null hypothesis: $H_0 : \mu = \mu_0$

- Test statistic: $t_{n-1} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$

- P-value:

If $H_1 : \mu \neq \mu_0$, then $p\text{-value} = 2 \times P(T_{n-1} > |t_{n-1}|)$.

If $H_1 : \mu > \mu_0$, then $p\text{-value} = P(T_{n-1} > t_{n-1})$.

If $H_1 : \mu < \mu_0$, then $p\text{-value} = P(T_{n-1} < t_{n-1})$.

- Decision: At a significance level of 0.05, we reject H_0 if $p\text{-value} < 0.05$.

$t\text{-test}(\text{data}, \mu = \mu_0, \text{alternative} = \text{"greater"})$
less
two.sided

Assumptions for t Methods

- ① normality
- ② independent samples

We can make inferences about means since the sample mean has an approximately Normal distribution (Central Limit Theorem).

In practice we do not know σ so have to use the t distribution with the sample standard deviation instead.

However, we still need to assume the sample mean is approximately Normal for p -values and confidence intervals to be correct.

Assumptions for t Methods

Some recommendations for how big n should be are:

- For small samples (roughly $n < 15$) only use the t methods if the data are **close to symmetric** and there are **no outliers**.
- For moderate samples ($15 \leq n \leq 40$) use the t methods as long as there are **no outliers** or **strong skewness** in the data.
- For large samples (roughly $n > 40$) use the t methods even in the presence of strong skewness, though **outliers** may still affect results.

Normal probability plots are a useful means for checking for skewness and outliers.

Errors and Power

	Decision	
	Retain	Reject
H_0 is true	Correct (1 - α)	<u>Type I Error</u> (α)
H_0 is false	<u>Type II Error</u> (β)	Correct (1 - β)